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T0 Projection onto Scramblon Physics: A
Quantitative Bridge between Geometric
Spacetime Correction and Quantum
Many-Body Chaos

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Abstract

We present a quantitative comparative analysis between the T0 Time-Mass Duality theory and scramblon physics, motivated by the work of Li, Zhou, Zhang et al. (Phys. Rev. Lett. **136**, 060403,

2026), who achieved the first experimental extraction of the quantum Lyapunov exponent in a many-body system. We show that the T0 damping factor $\mathcal{D}(N) = \exp(-\xi \ln(N)/\Delta f)$ provides a complementary, parameter-free description operating at the same order of magnitude ($\sim 10^{-4}$) as the scramblon corrections. The central testable prediction is a $\ln(N)$ -scaling residuum after complete scramblon error correction, which qualitatively differs from standard error models ($1/N$ or e^{-N}). Consistency is demonstrated across different platforms: NMR spin systems (adamantane, $N = 16$), IBM Brisbane (73 qubits), and IBM Sherbrooke (127 qubits). Both approaches – scramblons for dynamics, T0 for geometry – are complementary, not contradictory.

.1 Introduction

Motivation

Li et al. [1] have achieved a milestone in experimental quantum chaos research with their work *Error-resilient Reversal of Quantum Chaotic Dynamics Enabled by Scramblons*: the first extraction of the quantum Lyapunov exponent in a many-body system with macroscopic degrees of freedom. Using solid-state nuclear magnetic resonance (NMR) on adamantane ($C_{10}H_{16}$), they measure the out-of-time-ordered correlator (OTOC) and validate scramblon theory as a universal framework for information scrambling.

This document investigates how the T0 Time-Mass Duality theory [2] relates to these results. We show that the universal T0 correction factor – derived from fractal spacetime geometry with **zero free parameters** – provides a complementary description making consistent predictions at the same order of magnitude.

Core Question

The central question is not whether T0 *replaces* scramblon theory, but whether both frameworks describe different aspects of the same phenomenon:

- **Scramblon theory:** Dynamic description – how fast quantum chaos grows in a system
- **T0 theory:** Geometric description – what fundamental, system-size-dependent damping follows from fractal spacetime structure

Structure

Section .2 describes the experimental system. Section .3 derives the T0 projection. Section .4 compares both frameworks quantitatively. Section .5 demonstrates cross-platform consistency. Section .6 presents T0 predictions for the Lyapunov exponent. Section .7 formulates the testable prediction.

.2 The Experimental System

Adamantane NMR Setup

Li et al. use powdered adamantane ($C_{10}H_{16}$) in a 9.4 T magnetic field. The relevant system parameters are:

Table 1: Experimental parameters of the NMR system (Li et al.)

Parameter	Value	Significance
Sample	Adamantane ($C_{10}H_{16}$)	Plastic crystal
Spins per molecule	16	Protons (1H)
Magnetic field B_0	9.4 T	Superconducting
Larmor frequency	400.2 MHz	1H at 9.4 T
Dipolar coupling	~ 6.4 kHz	Intermolecular
Char. time τ_c	$\sim 156 \mu s$	$1/f_{\text{dipolar}}$

Each adamantane molecule contains 16 proton spins. Rapid thermal molecular motion averages out intramolecular interactions; the remaining intermolecular dipolar coupling (~ 6.4 kHz) serves as the dominant spin-spin interaction. The sample is placed in a uniform magnetic field that aligns the spin quantization axes.

Scramblon Theory: Summary

Scramblon theory [5, 6, 7] identifies *scramblons* as a novel type of collective excitation in quantum systems with all-to-all connectivity. Their key properties:

- Scramblons mediate the spread of quantum information, analogous to phonons mediating density fluctuations
- The scramblon propagator grows *exponentially* in time, dominating the late-time behavior of the OTOC
- In systems with holographic duality, scramblons in the boundary quantum theory correspond to gravitons in the bulk gravity

The central fitting ansatz of Li et al. takes the form:

$$F_{\alpha\gamma}(\phi, t) = f(e^{it}, a, b_\alpha, b_\gamma, \phi) \quad (1)$$

where $\varkappa$ is the Lyapunov exponent, a the imperfection coefficient, and b_α, b_γ are operator-dependent coefficients – a total of **4+ free parameters** per Hamiltonian.

.3 T0 Projection onto the NMR Number Space

The Universal Parameter and Its Projection

The fundamental T0 parameter $\xi = \frac{4}{30000} \approx 1.333 \times 10^{-4}$ is universal. How it manifests in a concrete physical system depends on the respective *number space* – the mathematical structure in which it operates.

Definition .3.1 (T0 Damping Factor for N -Spin Systems). For a system with N correlated spins, the T0 damping factor reads:

$$\mathcal{D}(N) = \exp\left(-\frac{\xi \ln(N)}{\Delta f}\right) = N^{-\xi/\Delta f} \quad (2)$$

with $\Delta f = 3 - \xi = 2.999867$ (fractal dimension) and **zero free parameters**.

Remark .3.2. The crucial point: ξ is not a fit quantity. It follows from T0 theory as a geometric parameter of fractal spacetime. The number space of each system – qubit register, spin ensemble, loop order – only determines *how* ξ enters the calculation.

Numerical Results

Table 2: T0 damping factor for various effective cluster sizes in the adamantane system

N (eff. spins)	$\mathcal{D}(N)$	$1 - \mathcal{D}(N)$	Correction (ppm)
16	0.99987678	1.232×10^{-4}	123.2
32	0.99984597	1.540×10^{-4}	154.0
64	0.99981517	1.848×10^{-4}	184.8
100	0.99979534	2.047×10^{-4}	204.7

The T0 correction is at the $\sim 10^{-4}$ level. Li et al. report that their experimental 95% confidence intervals are also at $\sim 10^{-4}$

(error bars are contained within the data markers). The T0 signature thus operates **exactly at the boundary of their measurement precision**.

.4 Quantitative Comparison: Scramblon Dynamics vs. T0 Geometry

Temporal Structure

Scramblon theory describes the OTOC as a time-dependent function with exponential growth:

$$F(t) \approx 1 - \frac{1}{N} \cdot e^{\varkappa t} \quad (\text{early times}) \quad (3)$$

where $\varkappa$ is the Lyapunov exponent, bounded by the Maldacena-Shenker-Stanford bound:

$$\varkappa \leq \frac{2\pi k_B T}{\hbar} \quad (4)$$

The T0 damping factor $\mathcal{D}(N)$ is, by contrast, *time-independent* – it describes a fundamental geometric correction, not dynamics.

Table 3: Time comparison: scramblon OTOC decay vs. static T0 correction ($N = 16$, $\varkappa \cdot \tau_c \approx 1$)

t/τ_c	OTOC scrambling	T0 correction	Ratio T0/Scr.
0.1	6.907×10^{-2}	1.232×10^{-4}	0.0018
0.5	1.030×10^{-1}	1.232×10^{-4}	0.0012
1.0	1.699×10^{-1}	1.232×10^{-4}	0.0007
2.0	4.618×10^{-1}	1.232×10^{-4}	0.0003
3.0	1.000 (saturated)	1.232×10^{-4}	–
5.0	1.000 (saturated)	1.232×10^{-4}	–

Remark .4.1 (Complementarity). Scramblon dynamics and T0 geometry operate on different levels:

- **Scramblons:** Describe *how fast* information spreads (time-dependent, exponential)
- **T0:** Describes *what fundamental limit* exists (time-independent, geometric)

At early times ($t \ll \tau_c$), the T0 correction dominates over the still-small scramblon signal. At late times, the exponential scramblon dynamics dominates. Both effects coexist.

Why Li et al. Do Not Separately Observe the T0 Effect

There are three reasons why the T0 correction is not identified as a separate effect in the data of Li et al.:

1. **Magnitude:** The T0 correction ($\sim 1.2 \times 10^{-4}$) lies exactly at the edge of the experimental confidence intervals ($\sim 10^{-4}$).
2. **Absorption in fits:** The scramblon fitting ansatz (Eq. 1) has 4+ free parameters that can readily absorb a static correction of this magnitude.
3. **No variation:** Li et al. did not systematically vary the effective cluster size N , which leaves the $\ln(N)$ scaling of the T0 correction invisible.

.5 Consistency Across Different Platforms

Same Formula, Different Number Spaces

A central feature of T0 theory: The same formula (Eq. 2), applied to different physical systems (= different number spaces), yields consistent corrections.

Table 4: T0 correction factor across different platforms

System	N	$\mathcal{D}(N)$	$1 - \mathcal{D}(N)$ (ppm)	CHSH (T0)
Adamantane NMR (molecule)	16	0.99987678	123.2	2.828079
Adamantane NMR (cluster)	64	0.99981517	184.8	2.827904
Superconducting (prediction)	73	0.99980932	190.7	2.827888
Superconducting (prediction)	127	0.99978472	215.3	2.827818

Cross-platform consistency of the formula

The transition from NMR spins ($N = 16$) to superconducting qubits ($N = 127$) represents a change of the entire physical system – different hardware, different couplings, different error sources. Yet the same single-line formula describes

the theoretical corrections consistently. The CHSH(T0) column contains *theoretical* prediction values, not measurements. Actual hardware measurements (Doc. 147 [3], IBM Heron r2, May 2026) give $S = 2.74$ – the predicted ξ effect ($\sim 10^{-5}$) lies far below the NISQ noise and is not resolvable with present hardware.

Hardware as Number Space

The effective ξ value extracted from experimental data depends on the hardware – not because ξ changes, but because the hardware itself is part of the number space:

Table 5: ξ extraction: Hardware convergence toward the theoretical value

System	N qubits	$\xi_{\text{fit}} (\times 10^{-4})$	$\xi_{\text{fit}}/\xi_{\text{base}}$	Excess
Theory	–	1.333	1.00	–
IBM Brisbane	73	2.29 ± 0.26	1.72 ± 0.19	72%
IBM Sherbrooke	127	1.37 ± 0.03	1.03 ± 0.02	3%

Better hardware converges toward the theoretical value because the number space of the hardware more closely approximates the ideal number space. The 127-qubit system maps the mathematical structure onto which ξ is projected more faithfully than the 73-qubit system.

.6 T0 Prediction for the Lyapunov Exponent

Modification by Fractal Geometry

Scramblon theory extracts a Lyapunov exponent $\varkappa$ from the experimental data. T0 predicts that this exponent should receive a system-size-dependent correction:

$$\varkappa_{\text{eff}} = \varkappa_{\text{bare}} \cdot N^{-\xi/\Delta f} \quad (5)$$

with the universal exponent $-\xi/\Delta f = -4.445 \times 10^{-5}$.

Table 6: Predicted T0 correction of the Lyapunov exponent

N (eff.)	$\mathcal{N}_{\text{eff}}/\mathcal{N}_{\text{bare}}$	Correction (ppm)
16	0.99987678	123.2
64	0.99981517	184.8
100	0.99979534	204.7
1000	0.99969302	307.0
10000	0.99959072	409.3

Remark .6.1. For typical NMR cluster sizes, the correction is small ($\sim 100\text{--}200$ ppm), but it grows systematically with $\ln(N)$. For future quantum simulators with $N > 10^4$ effective degrees of freedom, the effect could enter the measurable range.

.7 Testable Prediction

The $\ln(N)$ Residuum

Central Testable Prediction

After complete scramblon error correction (extrapolation to the error-free OTOC), a **residuum** should remain that scales **logarithmically with the cluster size N** :

$$R(N) = 1 - \text{OTOC}_{\text{corrected}}(N) \approx \frac{\xi \cdot \ln(N)}{\Delta f} \quad (6)$$

Numerical values:

$$R(16) \approx 1.23 \times 10^{-4} \quad (7)$$

$$R(100) \approx 2.05 \times 10^{-4} \quad (8)$$

This $\ln(N)$ scaling distinguishes T0 from all standard error models, which typically scale as $1/N$ or e^{-N} .

Experimental Test Protocol

For experimentalists who have access to the data of Li et al. or similar NMR scrambling measurements:

1. **Perform scramblon error correction** (as described in Li et al.) and extract the error-free OTOC for various effective cluster sizes N .
2. **Compute the residuum:** $R(N) = 1 - \text{OTOC}_{\text{corrected}}(N)$
3. **Plot $R(N)$ versus $\ln(N)$.** T0 predicts a linear relationship with slope $\xi/\Delta f \approx 4.44 \times 10^{-5}$.
4. **Cross-platform comparison:** Apply the same analysis on different hardware platforms (superconducting, ion trap, photonic). The residuum should be platform-dependent, but the $\ln(N)$ scaling universal.

The barrier to entry is minimal: The Zenodo raw data from Li et al. are publicly available (DOI: 10.5281/zenodo.16142455), and the T0 correction is a single line of code:

$$\mathcal{D}(N) = \exp\left(-\frac{4/30000 \cdot \ln(N)}{3 - 4/30000}\right) \quad (9)$$

.8 Method Comparison

Table 7: Systematic comparison: Scramblon theory vs. T0 projection

Property	Scramblon Theory (Li et al.)	T0 Projection
Foundation	Effective field theory	Geometric correction factor
Free parameters	4+ (fits per Hamiltonian)	0 (derived from ξ)
Expression	Multi-step fitting ansatz	$\mathcal{D}(N) = e^{-\xi \ln(N)/\Delta f}$
Describes	Dynamics (chaos rate)	Geometry (fundamental limit)
Time-dependent?	Yes (exponential)	No (static correction)
Platform	NMR-specific	Universal
Validated?	Yes (adamantane)	Yes (IBM 73/127 qubits)

.9 Conclusion

This analysis demonstrates that T0 theory and the scramblon physics of Li et al. provide **complementary, not contradictory** descriptions of the same phenomenon:

1. **Scramblons** describe the *dynamics*: how fast quantum chaos grows in a system. They require a complete effective field theory apparatus with multiple fit parameters.
2. **T0** describes the *geometry*: what fundamental, system-size-dependent damping follows from fractal spacetime structure. It requires a single universal parameter and one line of mathematics.
3. Both approaches are **experimentally grounded** – Li et al. in NMR measurements, T0 in IBM quantum processor data – and yield corrections at the same order of magnitude ($\sim 10^{-4}$).
4. A **testable prediction** has been identified: the $\ln(N)$ -scaling residuum after scramblon error correction. This prediction can be verified using publicly available raw data.

The analogy to historical parallels is apt: Just as Heisenberg's matrix mechanics and Schrödinger's wave mechanics provided different mathematical descriptions of the same physics, scramblons and T0 may represent different projections of a deeper structure. The experimental distinction lies in the scaling behavior – $\ln(N)$ for T0 vs. $1/N$ for standard error models – and is testable with current technology.

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